Comparing Anti-Forgetting Methods in Continual Learning

Mini-Project Report — Incremental Learning

Advanced Machine Learning — 2CS-SID — 2025/2026

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1. Introduction

Continual learning (also called incremental or lifelong learning) is the problem of training a neural network on a sequence of tasks without forgetting previously acquired knowledge. When a network is trained sequentially on multiple tasks using standard stochastic gradient descent, updates for new tasks overwrite the weights that encoded past tasks — a phenomenon known as **catastrophic forgetting** [1].

The objective of this project is to compare four strategies on a real-world benchmark: a baseline with no protection, and three anti-forgetting methods — Experience Replay, Learning without Forgetting (LwF), and Elastic Weight Consolidation (EWC). Performance is measured using average accuracy, backward transfer, forward transfer, and forgetting.

The chosen dataset is the **UCI Human Activity Recognition (HAR)** dataset, a real-world benchmark of smartphone accelerometer and gyroscope measurements. HAR was selected because it provides clearly separable activity classes that naturally map to sequential binary tasks, offers sufficient data volume for meaningful continual learning experiments, and is representative of a practical deployment scenario (wearable sensing) where continual learning is highly relevant.

2. Dataset & Experimental Setup

2.1. Dataset

The UCI HAR dataset [4] contains 561 time- and frequency-domain features extracted from accelerometer and gyroscope signals collected from 30 volunteers performing 6 activities: Walking, Walking Upstairs, Walking Downstairs, Sitting, Standing, and Laying. The dataset comprises 7352 training samples and 2947 test samples. All features were standardised to zero mean and unit variance using a `StandardScaler` fitted on the training set.

The dataset was split into **3 sequential binary tasks**:

Task	Classes	Train samples	Test samples
T1	Walking (0) vs. Walking Upstairs (1)	2308	927
T2	Walking Downstairs (2) vs. Sitting (3)	2534	1020
T3	Standing (4) vs. Laying (5)	2510	1000

2.2. Model Architecture

All methods share the same fully-connected architecture to ensure fair comparison:

Layer	Units	Activation
Input	561	—
Dense	256	ReLU
Dense	256	ReLU
Output	1	Sigmoid

The output layer uses a single sigmoid neuron (shared-head binary classification). Total trainable parameters: $561 \times 256 + 256 + 256 \times 256 + 256 + 256 \times 1 + 1 = 210{,}689$.

2.3. Methods Compared

- **Fine-tune** (baseline): sequential training with standard Adam, no forgetting protection — serves as lower-bound reference.
- **Experience Replay**: a FIFO memory buffer stores a fixed number of past samples; each training batch mixes current and replayed data.
- **LwF** (Li & Hoiem, 2016 [3]): knowledge distillation from a frozen copy of the previous model forces the new model to preserve old-task output distributions — no raw data is stored.
- **EWC** (Kirkpatrick et al., 2017 [2]): a quadratic penalty weighted by the diagonal Fisher information matrix penalises changes to weights that were important for past tasks.

2.4. Hyperparameters

Hyperparameter	Fine-tune	EWC	LwF	Replay
Learning rate	10^{-3}	10^{-3}	10^{-3}	10^{-3}
Epochs per task	5	5	5	5
λ	—	100	1.0	—
Temperature T	—	—	2.0	—
Buffer size	—	—	—	2000
Replay batch size	—	—	—	64
Fisher samples	—	500	—	—

3. Results

3.1. Accuracy Matrices

Table 1 shows the accuracy matrix $A_{i,j}$ (accuracy on task j immediately after training task i) for each method. The diagonal entries represent within-task accuracy at training time; below-diagonal entries reveal forgetting.

3.2. Summary Table

3.3. Metric Definitions

Let K be the total number of tasks and $A_{i,j}$ the accuracy on task j after training task i. The metrics are defined as:

Table 1: Accuracy matrices $A_{i,j}$ for all four methods. Bold values on the diagonal; forgetting visible below it.

Method	After	T1	T2	T3
Fine-tune	T1	**0.959**	0.621	0.481
	T2	0.514	**0.999**	0.502
	T3	0.513	0.514	**0.993**
Replay	T1	**0.965**	0.610	0.658
	T2	0.908	**0.987**	0.502
	T3	0.872	0.842	**0.990**
LwF	T1	**0.974**	0.420	0.574
	T2	0.645	**0.993**	0.509
	T3	0.560	0.538	**0.999**
EWC	T1	**0.970**	0.393	0.620
	T2	0.514	**1.000**	0.502
	T3	0.513	0.531	**0.993**

Table 2: Continual learning metrics for all methods. ↑ higher is better; ↓ lower is better. Best value per column in **bold**.

Method	Avg Accuracy ↑	Forgetting ↓	BWT ↑	FWT ↑
Fine-tune	0.6734	0.3103	−0.4654	+0.0618
EWC ($\lambda = 100$)	0.6792	0.3086	−0.4629	−0.0523
LwF ($\lambda = 1.0$)	0.6991	0.2897	−0.4346	−0.0353
Replay (cap. 2000)	**0.9011**	**0.0793**	**−0.1190**	**+0.0563**

$$\bar{A} = \frac{1}{K}\sum_{j=1}^{K} A_{K,j} \tag{1}$$

$$F_j = \max_{i \le K} A_{i,j} - A_{K,j} \tag{2}$$

$$\overline{F} = \frac{1}{K-1}\sum_{j=1}^{K-1} F_j \tag{3}$$

$$\mathrm{BWT} = \frac{1}{K-1}\sum_{j=1}^{K-1}\big(A_{K,j} - A_{j,j}\big) \tag{4}$$

$$\mathrm{FWT} = \frac{1}{K-1}\sum_{j=2}^{K}\big(A_{j-1,j} - b_j\big) \tag{5}$$

where $b_j = 0.5$ is the random-baseline accuracy for binary tasks.

3.4. Final Accuracy per Task

Figure 1 shows the last row of A for each method — the final accuracy on each task after all three tasks have been trained.

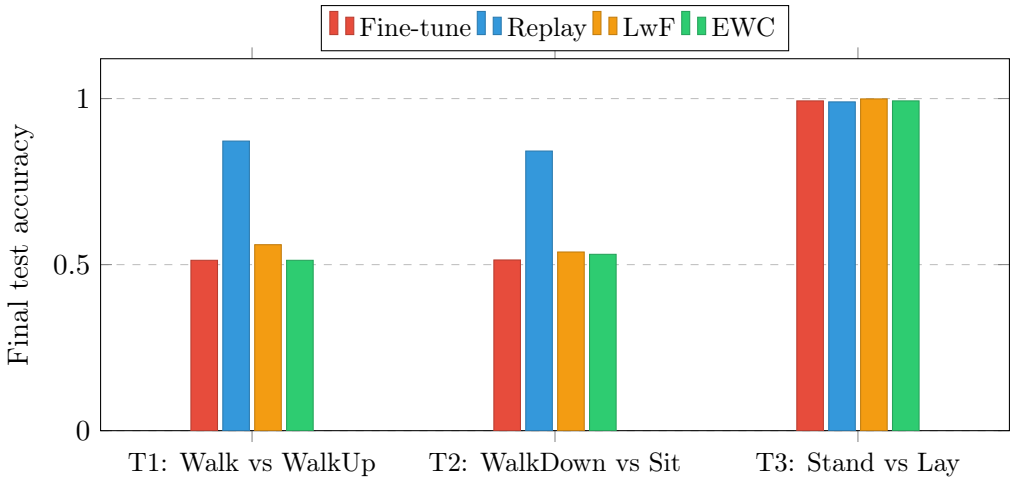

Figure 1: Final accuracy on each task after training all three tasks. Replay is the only method that retains meaningful accuracy on T1 and T2.

4. Analysis & Discussion

Best accuracy/forgetting trade-off. Experience Replay achieves by far the best trade-off: average accuracy of 0.901 (vs. 0.673–0.699 for all other methods) and forgetting of 0.079 (vs. 0.290–0.310). After all three tasks, per-task accuracies remain at 0.872 / 0.842 / 0.990 — demonstrating genuine retention of past knowledge alongside high current-task performance.

Expected ordering and surprises. We obtain $\bar{F}_{\text{fine-tune}} \geq \bar{F}_{\text{EWC}} \geq \bar{F}_{\text{LwF}} \gg \bar{F}_{\text{replay}}$. Replay clearly outperforms the rest ($\bar{F} = 0.079$), consistent with expectations. However, **EWC failed entirely** ($\bar{F} = 0.309$, essentially matching the baseline), and **LwF provided only marginal improvement** ($\bar{F} = 0.290$).

EWC failure analysis. The flat performance of EWC across $\lambda \in \{1, 10, 100, 500, 2000\}$ (average accuracy consistently ≈ 0.68) confirms the failure is structural rather than a tuning problem. Two causes are identified: (1) the shared single sigmoid output head changes semantic meaning between tasks — EWC's Fisher penalty can anchor hidden-layer weights but cannot resolve this output-level redefinition; (2) the HAR task pairs require largely non-overlapping representations, so the weights critical for old tasks are precisely the ones the new task must modify. A multi-head architecture would resolve issue (1).

LwF weakness. LwF's distillation signal is only useful when old and new tasks share input-space structure. The HAR task pairs are semantically dissimilar (walking dynamics vs. static postures), so the frozen old model's soft predictions on new-task data carry weak information about old-task behaviour. This results in a distillation loss that barely constrains forgetting: T1 accuracy still collapsed from 0.974 after T1 to 0.560 after T3, only marginally better than the baseline (0.513).

Influence of dataset choice. The high task dissimilarity of the HAR benchmark explains why Replay alone succeeded: it is the only method that explicitly re-presents old data, requiring no assumption about task similarity. Methods relying on implicit knowledge transfer (LwF) or weight-importance estimation (EWC) both require that tasks share representational structure, an assumption violated here.

Forward Transfer. FWT is near-zero for all methods (range: -0.052 to $+0.062$), confirming that past HAR tasks provide negligible zero-shot benefit to future tasks. Notably, both LwF

and EWC exhibit slightly negative FWT, suggesting their regularisation constraints modestly suppress the model's initial performance on future tasks.

Limitations. This experiment was run with a single random seed, which limits statistical reliability. The shared-head architecture is incompatible with EWC and disadvantages LwF. The small replay buffer (2000 samples across 3 tasks) limits coverage; the buffer-size sweep showed forgetting continued to decrease up to capacity 600–1000, suggesting further gains are possible. Finally, only 3 tasks are tested; longer sequences would likely amplify the performance gap between methods.

5. Conclusion

This project compared four continual learning strategies on the HAR dataset. Three key findings emerge: (1) **Experience Replay** is highly effective even with a modest buffer, reducing forgetting by 4× relative to fine-tuning and achieving 0.901 average accuracy; (2) **EWC and LwF both failed** to provide meaningful protection — EWC due to architectural incompatibility with the shared output head, LwF due to insufficient task similarity for effective distillation; (3) **high task dissimilarity** is the dominant factor shaping results, favouring data-driven methods (Replay) over regularisation-based ones. A natural next step would be to adopt a **multi-head architecture** and test on a longer task sequence with multiple random seeds to obtain statistically reliable comparisons.

References

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